

SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 03.1

Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB.

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Aim

To design Butterworth and Chebyshev Type-I band-pass IIR filters and compare their magnitude, phase, group-delay, impulse-response, and stability characteristics.

Procedure / Calculations

Normalize the band-pass edges, determine orders with `buttord()` and `cheb1ord()`, design both filters, plot their responses, compare group delay and impulse response, and check pole magnitudes.

MATLAB Program

```
clc; clear; close all;
% Specifications
Fs=2000; Rp=1; Rs=40;
Wp=[300 600]/(Fs/2);
Ws=[200 700]/(Fs/2);

% Butterworth Filter
[n1,W1]=buttord(Wp,Ws,Rp,Rs);
[b1,a1]=butter(n1,W1,'bandpass');

% Chebyshev Type-I Filter
[n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);
[b2,a2]=cheby1(n2,Rp,W2,'bandpass');

% Magnitude Response
figure;
freqz(b1,a1); title('Butterworth Filter');

figure;
freqz(b2,a2); title('Chebyshev Type-I Filter');

% Group Delay
figure;
grpdelay(b1,a1); hold on;
grpdelay(b2,a2);
legend('Butterworth','Chebyshev-I');

% Pole-Zero Plot
figure;
subplot(121),zplane(b1,a1),title('Butterworth')
subplot(122),zplane(b2,a2),title('Chebyshev-I')

% Impulse Response
figure;
subplot(211),impz(b1,a1),title('Butterworth Impulse')
subplot(212),impz(b2,a2),title('Chebyshev Impulse')

% Pole Magnitude
figure;
subplot(121),stem(abs(roots(a1)),'filled')
title('Butterworth Pole Magnitude')
subplot(122),stem(abs(roots(a2)),'filled')
title('Chebyshev Pole Magnitude')

disp(['Butterworth Order = ',num2str(n1)])
disp(['Chebyshev Order = ',num2str(n2)])
```

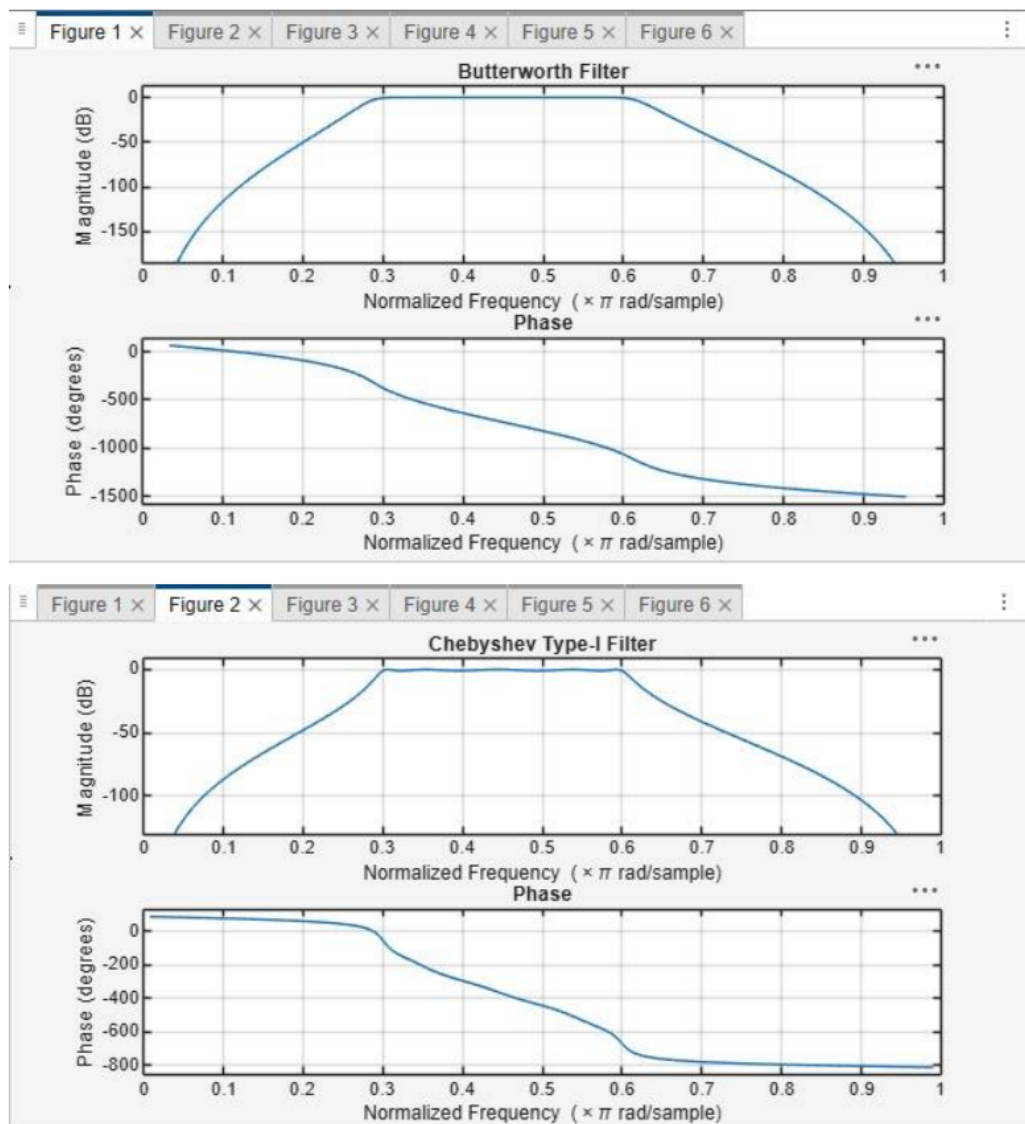
Output / Analysis

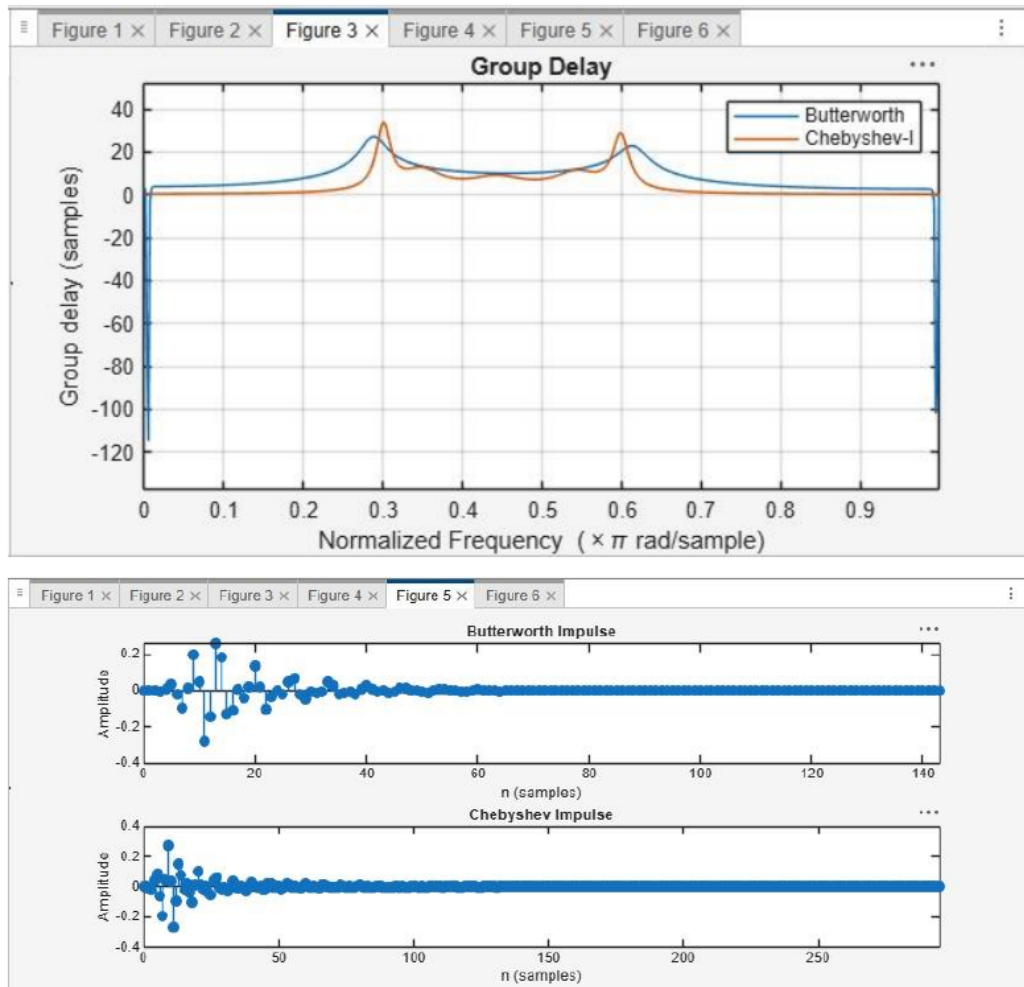
The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

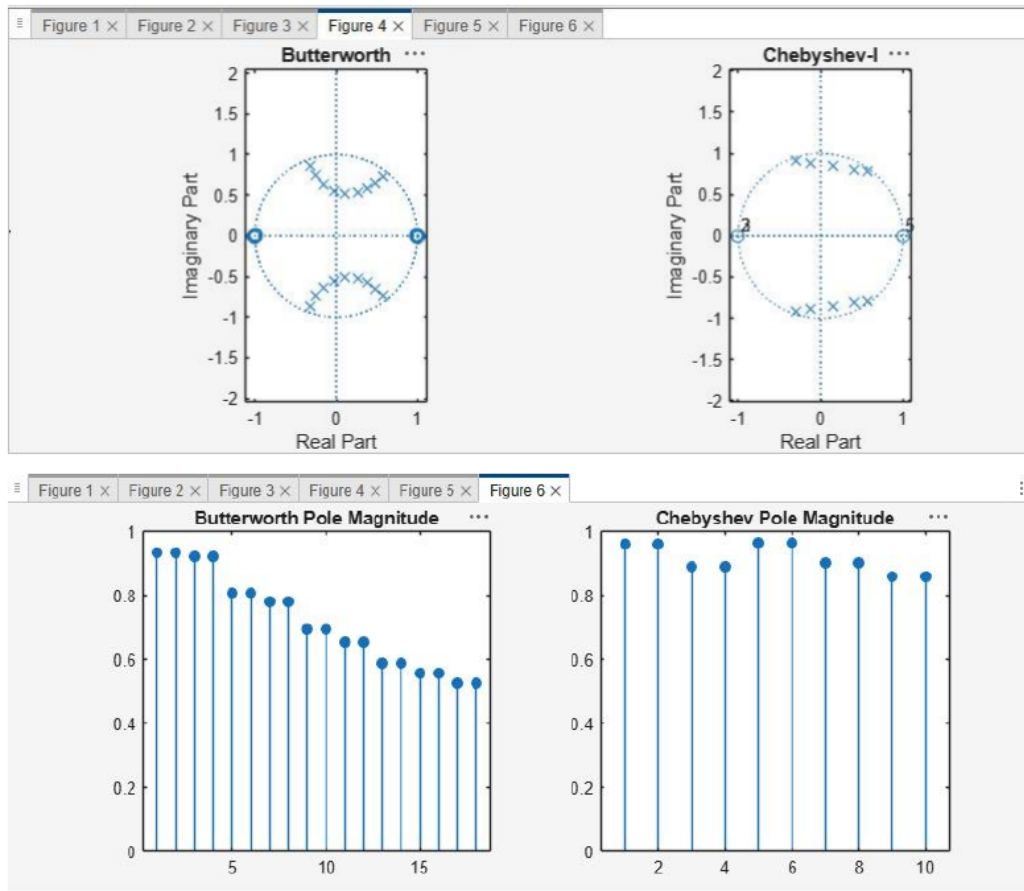
Output:

Butterworth Order = 9

Chebyshev Order = 5







Results:

1. Butterworth and Chebyshev Type-I IIR Band-Pass Filters were successfully designed and implemented in MATLAB.
2. The minimum filter orders and cutoff frequencies were obtained for both filters.
3. The frequency responses were analyzed and compared; the Chebyshev filter provided a sharper transition, while the Butterworth filter showed a smoother response.
4. Pole-zero analysis confirmed that both filters are stable and satisfy the given design specifications.

Practical Question – 1

Design and compare **Butterworth and Chebyshev Type-I IIR Band-Pass Filters** in MATLAB for the following specifications:

- Sampling Frequency = 10 kHz
- Passband = 1.5 kHz to 3 kHz
- Stopband = 1.2 kHz and 3.3 kHz
- Passband Ripple = 1 dB
- Stopband Attenuation = 45 dB

Tasks:

1. Determine the minimum filter order and cutoff frequencies.
2. Design both Band-Pass Filters and obtain the filter coefficients.
3. Compare the magnitude and phase responses.
4. Generate pole-zero plots and verify filter stability.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question -2

Design and compare **Butterworth and Chebyshev Type-I IIR Band-Pass Filters** in MATLAB for the following specifications:

- Sampling Frequency = 12 kHz
- Passband = 2 kHz to 4 kHz
- Stopband = 1.6 kHz and 4.5 kHz
- Passband Ripple = 0.5 dB
- Stopband Attenuation = 50 dB

Tasks:

1. Calculate the minimum filter order and cutoff frequencies.
2. Implement the filters using MATLAB and obtain the transfer function coefficients.
3. Analyze and compare the magnitude, phase, and group delay responses.
4. Plot the pole-zero diagrams and evaluate the stability and filtering performance.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

Both band-pass filters were successfully designed. The manual reports Butterworth order 9 and Chebyshev Type-I order 5; the plots show a sharper Chebyshev transition and stable pole locations.

Student Details

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